Fog Computing for Intelligent and Sustainable IoT: Architectures, Optimization, Datasets, and Open Challenges

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***Abstract*—The limitations of centralized cloud computing in IoT applications, which require low latency, high bandwidth, and mobility, have been demonstrated through the growth of the IoT ecosystem. Fog computing eliminates these limitations by distributing computing, storage, and networking closer to data sources across the edge-cloud continuum. This survey summarizes recent work on fog computing in terms of architectural design, resource management and scheduling; AI-based optimization; security/privacy issues; and energy efficiency. The survey also shows that fog systems can help reduce latency and bandwidth for real-time analytics in intelligent cities, healthcare, vehicle-to-vehicle networks, and industrial IoT.**

***Keywords*—Fog computing, IoT, Edge–Fog–Cloud continuum, AI-driven optimization, Resource management, green computing.**

I. Introduction

The emergence of IoT devices has led to an explosive increase in data volumes at the network edge. Despite the scalable storage and processing offered by CC, its centralized nature presents a harsh penalty to the IoT system, such as high latency, high bandwidth usage, high energy consumption, and privacy risks, especially in time-sensitive and data-intensive applications [5], [10], [11]. Such constraints render traditional cloud-based systems ineffective for new real-time services that demand low latency, high availability, and fast responsiveness. To address these network issues, FC has emerged as a distributed paradigm that brings data sources closer to computation, storage, and networking capabilities using geographically dispersed nodes. Compared with cloud models, the FC model offers the benefits of proximity-conscious processing, context-aware processing, and mobility, thereby supporting localised and timely decision-making. This minimizes the use of remote cloud centers and significantly enhances the quality of service (QoS), with an insignificant communication overhead and reduced response time [6].

The applications of FC are numerous in scenarios where latency is a critical consideration, such as smart healthcare, smart transportation, industrial automation, and smart cities [1], [16]. In this context, FC operates on data at the edge (i.e., gateways, roadside units, or local micro-servers), thereby enabling quick decision-making even in dynamic, mobile environments while alleviating network congestion and eliminating the need for a network connection.

FC can be structurally characterized by features such as low-latency processing, real-time analytics, location awareness, heterogeneity, and mobility support. Through abstractions such as Fog-as-a-Service (FaaS), Sensing-as-a-Service (SaaS), and Computing-as-a-Service, it extends cloud service models to the edge, providing flexible, on-demand access to resources. On the architectural level, fog systems usually take hierarchical, mesh (distributed), or hybrid forms, with each combining the features of scalability, resilience, and management complexity differently [7].

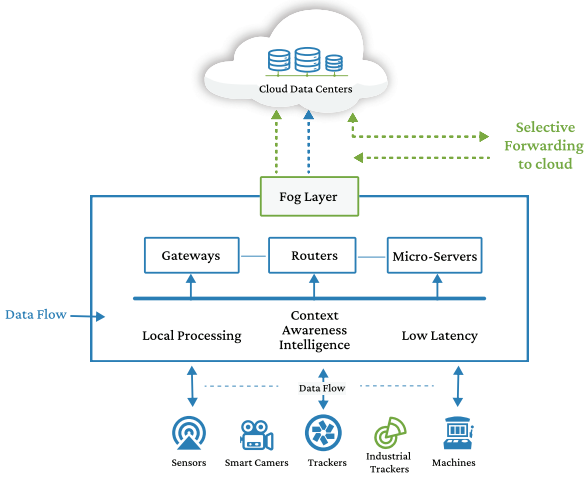


Fig. 1. Fog Computing Architecture

The three-layer fog architecture in Figure 1 represents the lowest level system of the network, composed of a vertical hierarchy where data first comes from the IoT edge, then gets processed in real time at the middle fog layer for low latency, and finally, if necessary, is sent to the cloud for heavy computation.This survey provides an up-to-date overview of the state of the art in FC, bridging gaps in the previous

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literature. The main contributions of the paper are as follows:

- An elaborate taxonomy of incorporating scheduling, resource utilisation, AI-based improvements, and energy saving procedures, backed by comparative tables across researchers.
- Systematic review of architectures, techniques, applications, challenges, and future directions, where FC plays a part in the IoT and its interaction with AI and green computing.
- Invest in new directions, including AI-optimized energy management in various fog setups, where the suggested directions are quantum-AI hybrid models.
- Improved accessibility by means of comparisons, real-world examples and case studies (e.g., vehicular FC).

The organization of the paper is structured as follows: Section 2 discusses the FC architecture, service models, Computing-as-a-Service, security/privacy mechanisms, and the applications of smart healthcare/vehicular networks. In Section 3, the resource management and task scheduling are discussed using adaptive schemes. Section 4 presents AI-based methods, including reinforcement learning (RL), deep learning (DL), and federated learning, along with their pros and cons. Section 5 focuses on energy efficiency through energy-aware scheduling, green AI techniques, and renewable energy integration. Section 6 describes a methodology for fog-AI evaluation, a literature review, and healthcare-focused datasets. Section 7 addresses open challenges and future directions. Section 8 summarises FC's progress, highlighting its performance benefits over cloud systems and its contributions to next-generation IoT.

II. Fog Computing Architecture and Service

FC provides a scalable environment that bridges the gap between resource-constrained edge devices and powerful cloud data centers. In contrast to cloud-only solutions, it moves processing and storage to data sources, thereby enabling low-latency processing, context-aware intelligence, and scalable management [5], [10], [16].

A. Architectural Paradigms

FC architecture is usually divided into three layers (1) edge computing devices (sensors, actuators, mobile devices) that provide raw data; (2) FC devices (gateways, routers, access points, micro-servers) to process the data locally, store it temporarily, and filter it; and (3) cloud data centers to compute the data in large scale, store it, and coordinate it globally. This multi-layered architecture enables flexible, application-specific deployments to achieve optimal performance, scalability, and reliability.

Fog architectures are classified into three paradigms, and they are dependent on calculation and coordination:

- Hierarchical Data flows designed as edge to edge, then to (when necessary) cloud, simple to manage, scalable, and predictable control is the best way to provide the applications that require data aggregation and policy implementation with minimal latency.

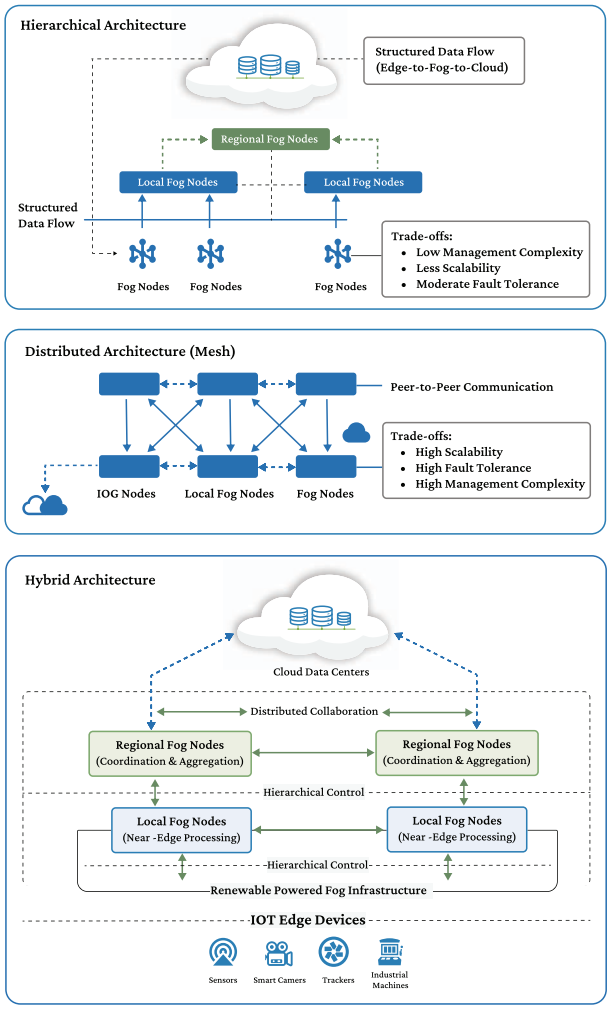


Fig. 2. Fog Architectural Paradigms

- Distributed (mesh): The fog nodes work in a peer-to-peer manner to make decentralized decisions and redundant routes, which make it highly resilient and fault-tolerant, which is appropriate in dynamic and robust environments.
- Hybrid: Combines hierarchical centralised oversight with distributed local collaboration, balancing scalability, performance, and reliability, best for large-scale, heterogeneous IoT systems.

The paradigms are presented in Figure 2, and the hierarchical (structured), distributed (decentralized mesh fault tolerance), and hybrid (balanced management and scalability) node connections are compared.

B. Service Models

FC extends cloud service models to the edge, utilising specialised abstractions that enable access to localised resources without requiring infrastructure management, making deployment more efficient. Some important models include FaaS, an on-demand model for computation, storage, and networking that reduces costs and enhances scalability [7]. This SaaS model involves localised data acquisition, prepro-

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cessing, and aggregation to minimise communication, ensure privacy protection, and facilitate real-time sensor analytics. The Computing-as-a-Service model, on the other hand, enables low-latency edge execution, allowing for near-time decision-making in monitoring and control systems. Together, these provide cloud-like functionality at the edge across various IoT applications [1], [11]. Security and privacy are vital, as distributed, heterogeneous nodes in open environments pose growing risks, including unauthorised access, interruptions, DoS attacks, and node intrusion. Effective mechanisms include AI-based intrusion detection systems for real-time anomaly detection against evolving attacks, and trust management frameworks, often integrated with blockchain, for decentralised reliability assessment, tamper-resistant records, and reduced dependence on central authorities [2], [30].

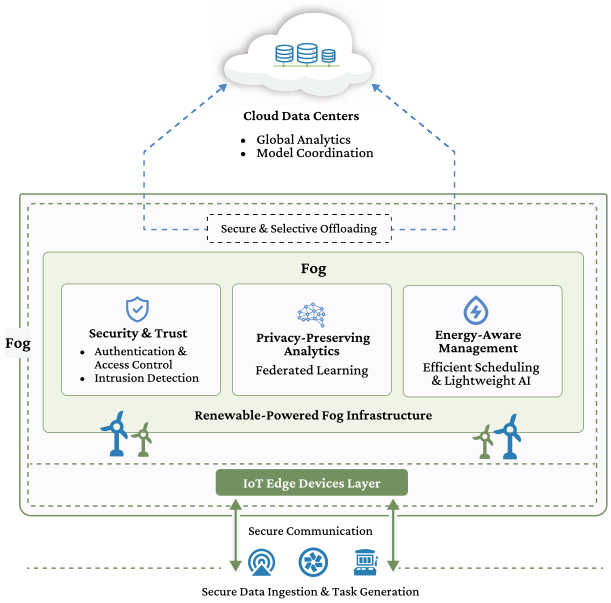


Fig. 3. Security, Privacy, and Energy-Aware Fog Computing Model

The security, privacy, and energy-aware model shown in Figure 3 integrates the safeguards such as intrusion detection and federated learning for privacy-preserving analytics, while promoting sustainability through lightweight AI models and energy-efficient scheduling to minimise environmental impact. Secure authentication and access control are the primary methods for preventing unauthorised access, leveraging lightweight protocols, role-based or attribute-based mechanisms, and context-aware authorisation to minimise overhead on resource-limited fog nodes [4]. Privacy preservation is very important in sensitive data applications; FC allows local processing to keep the data exposure to a minimum, which is further strengthened by federated learning where nodes only share model updates during collaborative training and no raw data is transferred and also by lightweight cryptography, key management, and anonymization that facilitate confidential, low-latency analytics [3], [22]. To sum up, fog security and privacy entail multi-layered approaches that incorporate detection, trust, authentication, and privacy techniques to build trustworthy IoT systems. FC excels in applications that require low latency, real-time responses, and efficient bandwidth. It is instrumental in smart healthcare, where it supports local analysis of data from wearable sensors for instantaneous anomaly detection and alert, thereby increasing responsiveness and privacy.

III. Resource Management and Task Scheduling

The effectiveness of resource management and task scheduling largely determines the success of FC, which is characterized by heterogeneous hardware, limited resources, dynamic networks, and variable workloads.

A. Scheduling Strategies

Examples of scheduling strategies include heuristic and metaheuristic algorithms (e.g., genetic algorithms, particle swarm optimisation, ant colony optimisation), which dynamically assign tasks based on processing power, network delay, workload, and energy to reduce latency and costs in large-scale, heterogeneous setups [14], [28]. The fault-tolerant techniques, on the other hand, raise the dependability level by means of process replication, migration, or checkpointing to handle node failures caused by hardware issues, connectivity loss, or energy depletion [24]. Multi-objective optimisation balances the opposing QoS goals of delay, energy use, utilisation, and fairness, thereby providing solutions for diverse applications [15], [31]. Studies show fog-aware scheduling reduces end-to-end latency by 20%–40% compared to cloud-only models, while boosting resource utilisation and responsiveness, underscoring its importance for demanding IoT applications [8], [21].

IV. AI-Driven Fog Computing

To solve the challenges of FC, which include distributed control, heterogeneous resources, dynamic workloads, and strict latency demands, artificial intelligence (AI) is the main game changer. Static or rule-based methods are usually unsuccessful when dealing with such kinds of variability [14], [15]. AI systems at FC learn from experience, adapt to changes, and autonomously optimize performance, energy efficiency, and reliability [27], [29]. This is why AI has become indispensable for intelligent FC.

A. Reinforcement Learning and Deep Learning

Among AI methods, RL is the most significant, enabling adaptive, online decision-making in dynamic fog environments. RL helps fog nodes offload tasks, allocate resources, and balance load through trial-and-error interactions with the environment to maximise cumulative rewards. Deep RL takes this a step further by employing neural networks to handle complex state spaces and is therefore suitable for large-scale mobile deployments. Another move towards the decentralization of coordination of various nodes to support scale is Multi-agent RL. Research results indicate that RL-based solutions can achieve significant improvements in latency,

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energy consumption, and throughput compared to conventional methods. To maintain privacy, federated learning enables distributed nodes to learn collaboratively without sharing raw data, thereby preserving the accuracy and security of sensitive areas of human life, such as healthcare and smart cities [13], [20], [26]. The AI-driven resource management framework in Figure 4 adds intelligence to the fog layer, implementing RL and DL to independently decide on task offloading using a continuous feedback loop of performance measurements.

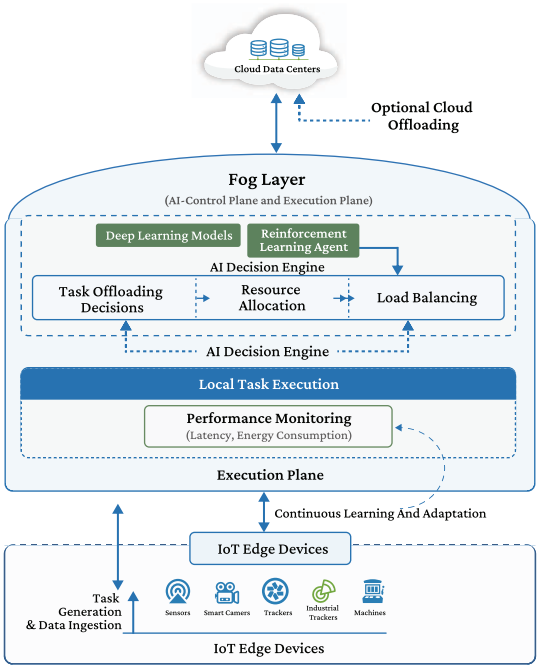


Fig. 4. AI-Driven Resource Management and Task Scheduling Framework

V. Energy Efficiency and Green Fog Computing

Energy efficiency and sustainability are the focal points of FC, driven by the rise in IoT deployments and the high energy requirements of digital infrastructure. Unlike cloud-centric models that increase consumption and emissions due to constant data transmission, FC encourages local processing, alleviates network traffic, and is designed to be energy-aware, in line with green objectives. The main instrument is energy-aware resource management and task scheduling, which allocate tasks based on a node's energy, workload, and communication costs in a manner that supports local processing and minimises offloading [1]. Green AI methods such as model pruning, quantisation, and lightweight architectures are designed to reduce computational requirements on resource-constrained nodes while maintaining accuracy. Moreover, the use of renewable energy sources (e.g., solar, wind) and energy-harvesting, together with adaptive management, allows for reduced dependence on the grid and smaller carbon footprints, especially in off-grid locations [17], [25]. Green AI techniques such as model pruning, quantisation, and lightweight architectures lower computational demands on constrained nodes while preserving accuracy. Additionally, integrating renewable energy sources (e.g., solar and wind) and energy harvesting with adaptive management reduces grid dependence and carbon footprints, especially in remote settings [21]. Research results indicate that fog processing can yield energy savings of 15%-50% compared to cloud-only methods, with savings varying by application and configuration; thus, fog plays an important role in the realisation of sustainable, high-performance IoT systems.

VI. Dataset Overview and Methodology

This chapter describes datasets and methods used in a comprehensive survey of FC combined with AI. FC, as a distributed extension of CC, processes data closer to IoT devices and sources, thereby reducing latency and enabling real-time decision-making. The dataset includes research articles on FC from 2015 to 2026 (with selected papers up to 2025) that focus on fog architecture, resource management, applications, and AI enhancements. The articles were systematically selected and reviewed to extract methodologies, applications, and datasets, and to make the survey friendly to non-experts. The integration of AI is the main driver of FC's growth, enabling intelligent edge processing and decision-making. Among the techniques are large language models (LLMs) for local medical data interpretation to preserve privacy [31], genetic algorithms for workflow optimization under deadlines, predictive fault tolerance, and energy-efficient green computing. On the one hand, AI applications include the SpezialLLM framework for privacy-focused healthcare AI in fog [26], AI for secure NFT/blockchain evidence systems [12], aand AI-enhanced key management in hybrid cloud-fog schemes. This dataset encompasses a variety of data types: simulated (e.g., synthetic data for homomorphic encryption testing in fog), real-world (e.g., healthcare records, supply chain RFID logs), and benchmark (e.g., security and key management data for fog-cloud systems). They support validation in scenarios such as multi-cloud energy optimization, security models, and health monitoring [21]. Although the dataset is invaluable for demonstrating practical relevance and data diversity, it is limited in size due to constraints on research compilation and operational scale.

Table I lists the eight unique datasets that have been used to measure the effectiveness of decentralized, fog-based architectures in the healthcare domain. These datasets include those related to SpeziLLM and the Stanford Biodesign Digital Health GitHub repositories. Some key examples include the VeReMi extension (2020), which provides about 3.15 million simulated vehicle message logs to detect attacks in vehicular networks. Helib-related plaintext vectors (from 2014 onwards): These are 10248192 slot vectors for homomorphic encryption in secure health computations. LLMonFHIR (2024 beta): It uses 520 simulated FHIR health records for LLM-based querying. HealthGPT (2024 MVP): It has incorporated 14 days of fitness tracker data for health coaching. OpenTSLM (2023-2025) focuses on medical multivariate time-series patterns. Pediatric Apple Watch Study (2023): It collects children's wearable log data. Engage-HF (2024): It is used for tracking

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TABLE I
DATASET SUMMARY

Dataset	Modality	Sample Size	Key Features
VeReMi Extension 2020	Simulation logs, GPS/BSM messages	≈3.15 million records (3,146,076 messages)	Vehicular misbehaviours (DoS, location spoofing); realistic sensor error models
HElib Plaintext Vectors 2014	Structured numerical (polynomial ring slots)	Scalable (approx. 1024–8192 slots per vector)	SIMD slot-packing, noise-based security, automorphisms, GHS optimizations for routing and linear algebra
LLMonFHIR Data 2024 (Beta)	Structured text (FHIR JSON)	Estimated 5–20 synthetic patients	Privacy-safe synthetic records; multilingual support; conversational explanations
HealthGPT Data 2024 (MVP)	Numerical and text time-series (HealthKit)	Estimated 14-day records; 5–20 synthetic users	Multiple health metrics (steps, sleep); local/fog extensibility; natural language queries
OpenTSLM Data 2023	Text and numerical time-series (multivariate medical)	Variable synthetic datasets	Reasoning over multivariate medical time-series; open-source modeling support
Pediatric Apple Watch Study Data 2023	Numerical and text time-series (Apple Watch)	Variable study participants	Wearable-based data collection; pediatric health monitoring
ENGAGE-HF Data 2024	Numerical and text health metrics	Variable patient data	Heart failure monitoring; integration with the Spezi framework
MyHeartCounts Data 2023	Numerical and text time-series (health metrics)	Variable participant data	Comprehensive cardiovascular health tracking; Spezi-based framework

heart failure metrics. MyHeartCounts (2023) is used to collect variable cardiovascular logs from participants. These datasets primarily use text and numerical formats, enabling privacy-preserving, real-time experiments with fog-integrated LLMs to test latency and security.

VII. OPEN CHALLENGES AND FUTURE DIRECTIONS

Significant advances have been made in FC architectures, intelligent resource management, and AI optimization. However, key issues must be resolved before the technology can be widely used and sustained over time. One major problem is scalability. Future IoT systems are expected to consist of millions of heterogeneous mobile fog nodes. Therefore, it is essential to develop efficient hierarchical mechanisms for scheduling and orchestration coordination that combine local autonomy with global stability. These mechanisms will also avoid the shortcomings of purely centralised or decentralised approaches. On the other hand, standardisation and interoperability are crucial, as the diversity of hardware, protocols, and proprietary frameworks hinders integration, application portability, and cross-layer (edge-fog-cloud) coordination. Thus, a vendor-agnostic ecosystem is needed that includes unified models, open APIs, and standardised abstractions.

TABLE II
COMPARISON OF CLOUD, EDGE, AND FOG COMPUTING

Ref	Parameter	CC	EC	FC
[1]	Computing Location	Centralised data centres	At or very close to the data source	Intermediate layer between edge devices and cloud
[2]	Latency	High	Very low	Low
[3]	Bandwidth Usage	High due to continuous data transmission	Minimal	Reduced through local and regional processing
[4]	Scalability	High (centralised scaling)	Limited	High (distributed scalability)
[5]	Mobility Support	Limited	Limited	Strong support
[6]	Context Awareness	Low	High	High
[7]	Resource Availability	Abundant	Highly constrained	Moderately constrained
[8]	Energy Efficiency	Low for IoT workloads	Moderate	High with energy-aware scheduling
[9]	Security Exposure	Centralised attack surface	Device-level vulnerabilities	Distributed but manageable
[10]	Privacy Preservation	Low (raw data transmission)	High	High with localised analytics
[11]	Fault Tolerance	High (data centre level)	Low	High through distributed fog nodes
[12]	AI Deployment Feasibility	High (resource-rich environment)	Limited	High using lightweight AI models
[13]	Typical Applications	Big data analytics, large-scale storage	Sensor-level control and actuation	Real-time IoT, smart cities, healthcare

AI intelligence also poses challenges for energy sustainability [9], [23]. The increased adaptability brought about by AI also entails greater computational overhead on resource-constrained nodes. Therefore, lightweight, adaptive models are needed that can dynamically adjust their complexity based on available energy and workload. Additionally, integration with emerging networks (5G/6G) and A-IoT requires implementing new mechanisms for network slicing, distributed learning, and trust management [19]. Future steps include hybrid fog-cloud intelligence with dynamic decision sharing, explainable AI for transparency in critical applications, and carbon-aware scheduling to prioritize environmental sustainability [7], [18], [20].

VIII. CONCLUSION

Fog computing is a recent significant paradigm for distributed systems that bridges the limited resources available at the edge of computer networks and centralized clouds. It has been proposed as a solution to several limitations posed by the cloud, such as latency, inefficient bandwidth use, poor energy

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utilization, and data privacy, as well as providing a foundation for developing responsive, context-aware Internet of Things (IoT) applications. This survey provides a general overview of key concepts related to fog computing, including architecture, service models, resource management, AI-oriented optimization, security/privacy, applications, and energy-efficient design. Based on an analysis of the reviewed literature, fog computing is more effective than cloud-based solutions in reducing latency, saving bandwidth, and improving real-time performance. However, challenges remain related to scalability, interoperability, energy-aware AI, and the integration of future networks; however, progress has been made in these areas. Future fog computing developments will provide the basis for building a new platform for next-generation distributed systems using hybrid fog and cloud intelligence, trusted AI, standardization of models, and sustainable optimization.

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